

High-Extinction-Ratio Integrated Polarizer Based on a Subwavelength Grating Bent Waveguide

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Abstract Silicon photonic devices show markedly different behaviors for TE and TM polarized modes, making polarization management critical. We experimentally demonstrate a TE-pass polarizer based on subwavelength-waveguide cut-off engineering and bend-assisted filtering, experimentally achieving an extinction ratio of 28 dB and insertion losses of -0.22 dB. ©2026 The Authors

Introduction

Silicon-on-insulator (SOI) is a widely exploited platform in integrated photonics since it offers strong optical confinement, compact device dimensions, and compatibility with mature CMOS fabrication processes [1]. However, submicron SOI waveguides exhibit strong polarization dependence because the transverse-electric (TE) and transverse-magnetic (TM) modes differ in effective index, propagation loss, and cutoff condition. As a result, silicon photonic circuits are usually optimized for a single polarization, and any undesired orthogonal polarization component may degrade performance [2]. This issue is particularly relevant in telecommunication systems, for example in polarization-diversity receivers, where the two orthogonal polarization states must be separated and processed independently, thereby increasing circuit complexity and footprint [3]. In this context, integrated devices for polarization management are essential components. In particular, optical polarizers are designed to selectively allow only one polarization to propagate and act as filters to ensure polarization purity. Figures of merit for integrated polarizers include high polarization extinction ratio (PER), low insertion loss, broad operating bandwidth, and small footprint, thereby facilitating the implementation of compact and efficient polarization-controlled photonic systems. Several integrated polarizers have been reported on the SOI platform. Among them, photonic-crystal polarizers can provide compact and strongly selective devices, but their bandwidth remains limited [4]. Bragg-grating-based polarizers can also achieve high selectivity, yet the rejected TE polarization is reflected, which can be detrimental to system performance in some applications [5]. To mitigate this issue, alternative Bragg grating designs were introduced to convert the reflected TE mode into a higher-order mode, but the reported PER remains relatively low [6]. Other polarizers exploit polarization-dependent bending losses in curved

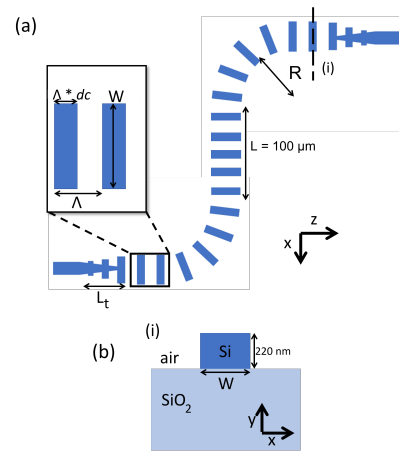


Fig. 1: Schematic of the TE-pass polarizer based on a bent subwavelength waveguide (SWG). (a) Device top view with the main geometrical parameters; (b) waveguide cross-section.

waveguides [7, 8]. Although effective, these designs require multiple cascaded bends to enhance polarization rejection, leading to a larger overall footprint. Another important class of integrated polarizers exploits the different cut-off conditions of the TE and TM modes. In these devices, the waveguide geometry is designed so that only one polarization remains guided and propagates, while the other reaches cut-off and is therefore filtered out. While this concept can be implemented with conventional waveguides, subwavelength grating waveguide (SWG) structures offer greater fabrication flexibility and improved control of the cut-off behavior [9, 10]. However, in both conventional and SWG implementations, partial reflection of the rejected polarization at the silicon substrate beneath the buried oxide (BOX) can couple light back into the waveguide and thus limit the achievable PER [9].

In this paper, we propose a TE-pass SOI polarizer that combines SWG cut-off engineering with polarization-dependent bending losses. Unlike straight cut-off-based designs, the proposed geometry uses bends not only to enhance TM rejection but also to suppress the recoupling of the rejected

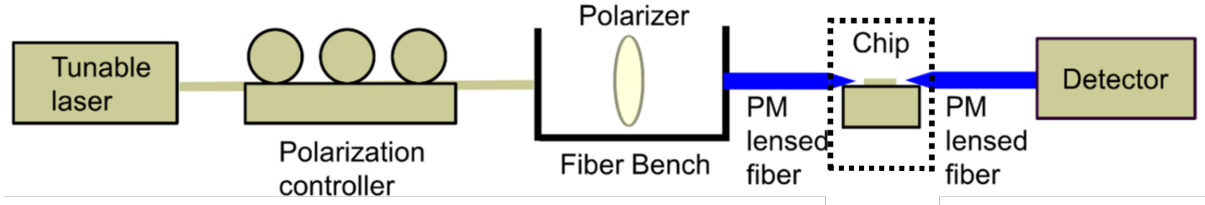


Fig. 2: Experimental setup for transmission measurements. PM : Polarization Maintaining

polarization after partial reflection from the silicon layer beneath the BOX. The combination of these two concepts allows the demonstration of an integrated polarizer with an experimental PER of about 28 dB, insertion loss below 1 dB, and a compact footprint, as the 90° bend confines the device extension in the initial propagation direction to only 30 μm .

Polarizer design

The proposed TE-pass polarizer is designed to operate around the central telecommunication wavelength of 1550 nm. As shown in Fig. 1, the polarizer is implemented on an SOI platform with a 220 nm-thick silicon layer, a 3 μm -thick buried oxide (BOX), and an air upper cladding. The device is composed of two curved subwavelength waveguides (SWG) sections connected by a straight SWG segment, and interfaced with the input and output wire waveguides through tapered transitions. Its geometry is defined by the waveguide width W , the SWG period Λ , the duty cycle dc , the bend radius R , and the taper length L_t . The values $W = 1100 \text{ nm}$ and $dc = 0.5$ are chosen because they provide a robust TE-pass operating point, for which only the fundamental TE-like Bloch mode is supported while the TM-like Bloch mode is cut off. Moreover, the 0.5 duty cycle preserves sufficiently large feature sizes, making the design more tolerant to fabrication [10]. In the present design, the period is fixed to $\Lambda = 200 \text{ nm}$ in order to remain well below the Bragg limit (i.e. $\Lambda < \lambda/2n_{eff}$), thereby reducing back-reflection and the associated insertion loss, while maintaining sufficient tolerance in the curved sections; smaller periods would also impose stricter fabrication constraints. This parameter set was further verified using the Meep/MPB suite [11], which confirmed single-mode TE-like propagation in the SWG region. The bend radius was fixed to $R = 11 \mu\text{m}$, as this value preserves low TE loss while enhancing TM attenuation through bending; the latter increases for smaller radii. Finally, the SWG-to-wire transition taper length was fixed to $L_t = 5 \mu\text{m}$, as this value provides a suitable trade-off between taper loss and device compactness [12].

The proposed TE-pass polarizer was then analyzed using three-dimensional finite-difference time-domain (3D-FDTD) simulations around the central wavelength of 1550 nm. To reduce the com-

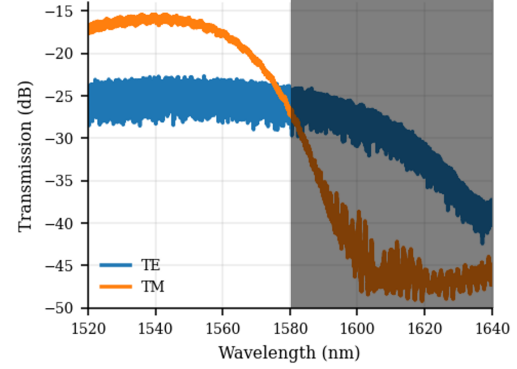


Fig. 3: Reference TE and TM transmission spectra measured on a straight ridge waveguide without polarizer.

putational cost, the simulated model includes only the input and output tapered transitions together with one curved SWG section, i.e., excluding the straight section and the second bend. The straight SWG section was investigated separately. In this work, the transmission coefficient T is defined from the ratio of output to input power and expressed in decibels as

$$T_{\text{TE, TM}} = 10 \log_{10} \left(\frac{P_{\text{out, TE, TM}}}{P_{\text{in, TE, TM}}} \right). \quad (1)$$

The polarization extinction ratio (PER) is then defined as the difference, in decibels, between the TE and TM transmissions. Simulations of the curved SWG section predict a TE insertion loss of about -0.07 dB and a PER of approximately 39 dB at 1550 nm. They also indicate that the PER remains above 35 dB over the 1500–1600 nm range, confirming the broadband response of the bent geometry. For comparison, a 20 μm -long straight SWG section with the same width, period, and duty cycle was also evaluated at 1550 nm. Although a 20 μm -long straight section is longer than the arc length of a single 90° bend with $R = 11 \mu\text{m}$, i.e. 17.28 μm , it provides a lower PER of only about 21 dB, while its TE loss remains smaller, around -0.017 dB . This comparison shows that the curved SWG section significantly enhances TM rejection, at the expense of a moderate increase in TE loss.

Experimental results

The polarizer was fabricated using an e-beam lithography based process.

The experimental setup used for the characterization of the polarizer is shown in Fig. 2. Light from

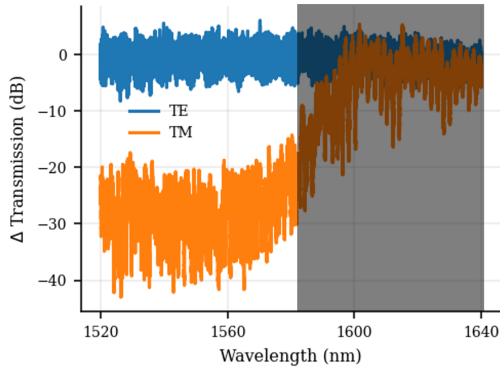


Fig. 4: Normalized measured TE and TM transmission of the TE-pass polarizer.

a tunable laser first passes through a polarization controller to align the input polarization state with that of the polarizer. It is then coupled into the chip through a polarization-maintaining (PM) lensed fiber. At the output, the transmitted signal is collected by a second PM lensed fiber and directed to a detector. This arrangement allows the transmission of the fabricated polarizer to be measured as a function of wavelength for the selected input polarization. At the chip facet, the optical field is coupled into the device using an inverse taper edge coupler with an input width of 350 nm. This width was selected because it provides the best overlap between the Gaussian mode delivered by the lensed fiber and the guided mode of the access waveguide, thereby maximizing the fiber-to-chip coupling. However, this narrow access waveguide also remains relatively close to the TM cut-off at long wavelengths; in our study, TM guidance is no longer supported beyond approximately 1580 nm. Increasing the access-waveguide width would move the structure further away from this cut-off and improve TM guidance at longer wavelengths, but at the expense of a reduced modal overlap with the incident fiber beam and, therefore, lower coupling efficiency. This trade-off is reflected in the reference measurements shown in Fig. 3 where the reference TE and TM spectra were measured on a straight ridge waveguide without the polarizer structure. These measurements are therefore not related to the filtering behavior itself, but are only used to characterize the coupling conditions of the setup. A strong degradation of the TM-coupled power is observed as the wavelength increases, whereas the TE reference remains significantly more stable over the same range. For the characterization of the polarizer, these reference spectra are used as normalization baseline.

Fig. 4 shows the normalized transmission spectra of the fabricated polarizer for TE and TM modes excitation. The TE response remains low-loss over the whole measured wavelength range, with an average transmission of about -0.22 dB around

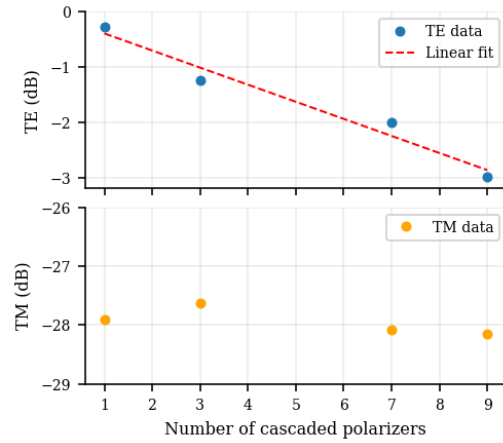


Fig. 5: Transmission as a function of the number of cascaded polarizers, averaged in the 1540-1560 nm wavelength range.

1550 nm. In contrast, the TM mode remains stably attenuated up to approximately 1580 nm. As a result, the device exhibits a stable PER close to 28 dB over this spectral range, confirming the TE-pass behavior of the fabricated polarizer. Beyond 1580 nm, the apparent evolution of the TM response is mainly attributed to the TM coupling inefficiency identified in the reference measurement. The low TE insertion loss further allows cascading several polarizers to probe the rejection limit of the device. Fig. 5 reports the transmission measured for different numbers of cascaded polarizers. The TE transmission decreases progressively, as expected from the accumulation of insertion loss, whereas the TM transmission remains limited to about -28 dB and does not decrease further. This behavior indicates that the measured PER is mainly constrained by the finite polarization extinction of the experimental setup rather than by the intrinsic rejection of the polarizer. The actual PER is therefore likely higher than the experimentally observed value.

Discussion and outlook

In this work, we demonstrated a TE-pass polarizer based on a bent SWG waveguide, with an experimental PER close to 28 dB, a TE insertion loss of about 0.22 dB, and stable operation over the 1520–1580 nm range. These results make the device a promising compact component for polarization control. The present characterization, however, remains influenced by the fiber-to-chip coupling conditions, particularly for TM excitation at longer wavelengths. The experimentally observed bandwidth is therefore likely narrower than the intrinsic operating range of the polarizer. Moreover, the saturation of the TM rejection around -28 dB suggests that the current PER is limited by the experimental setup rather than by the device itself. Further measurements in a free-space configuration will therefore be necessary to evaluate the intrinsic PER of the polarizer more accurately.

Acknowledgements

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